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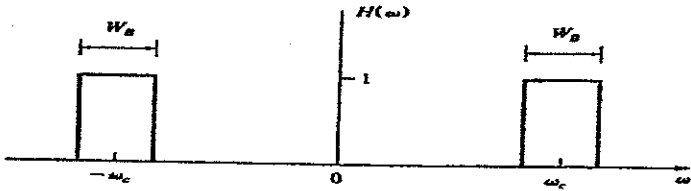
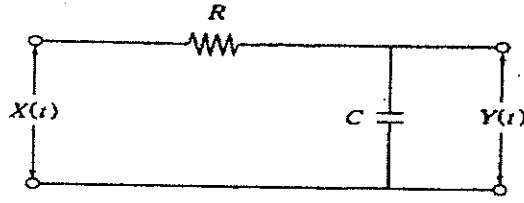
Degree: B. Tech. Semester: 3 - Course work
END-SEMESTER EXAMINATION, Nov-Dec 2023
 Course Title: Probability Theory and Stochastic Process
 Course Code: ECECC06/EIECC06

Time: 03 Hours

Max. Marks: 50

Note: - Attempt all the five questions. Missing data/ information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	(i) Two baskets, basket 1 and basket 2, contain, respectively, two white and one black ball, and one white and five black balls. One ball is transferred from basket 1 to basket 2 and then one ball is drawn from the latter. Suppose the drawn ball happens to be white. What is the probability that the transferred ball was black? (ii) Let X be the random variable with the following pdf. $f_X(X) = \frac{1}{2}e^{- x } \quad \text{for all } x$ Find CDF of Y if $Y=X^2$	2+3	1
1b	(i) Let X be a random variable whose probability density function is given by $f_X(x) = \begin{cases} e^{-2x} + \frac{1}{2}e^{-x}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$ Write down the moment generating function for X and use the same to compute the first and second moments of X.	3+2	1
1c	(i) Suppose X has the Moment Generating Function: $m_X(t) = (1 - 2t)^{-\frac{1}{2}} \quad \text{for } t < \frac{1}{2}$ Find the first and second moments of X. (ii) Define the law of total probability.	4+1	1
Q 2	Attempt any 2 parts of the following		
2a	Let (X, Y) be a bivariate r.v. with the joint pdf $f_{XY}(x, y) = \frac{x^2+y^2}{4\pi} e^{-(x^2+y^2)/2} \quad -\infty < x < \infty, -\infty < y < \infty,$ Are X and Y independent? Are X and Y uncorrelated? Justify your answer.	2+3	2
2b	(i) Let $X \sim \exp(\beta_1)$ and $Y \sim \exp(\beta_1)$ be independent RVs. (a) Find the joint density of X and Y (b) Calculate $P(X < Y)$ (c) If $Z = X - 2Y$, find $E(Z)$ and $Var(Z)$ (ii) The joint p.m.f of a bivariate r.v. (X, Y) is given by: $p_{XY}(x_i, y_j) = \begin{cases} kx_i^2 y_j, & x_i = 1, 2; y_j = 1, 2, 3 \\ 0, & \text{otherwise} \end{cases}$ Find the marginal pmf's of X and Y. Are X and Y independent?	3+2	2
2c	Suppose that X and Y are random variables such that $Var(X)=9$, $Var(Y)=4$, and $\rho_{X,Y}=1/6$. Determine $Var(X+Y)$ and $Var(X-3Y+4)$.	5	2
Q 3	Attempt any 2 parts of the following		
3a	State and prove central limit theorem	1+4	3
3b	Find the pdf $f_Y(y)$ for the following cases: (i) $Y = e^X$, and X is a uniform random variable over $(0, 1)$ (ii) $Y = X^2$, where X is a uniform random variable over $(-1, 2)$. Sketch the	2+3	3

3c	Consider a random variable X having probability density function $f_X(x) = 0$ for $x < 0$. For such random variable prove, the following inequality $P(X \geq k) = \frac{E[X]}{k} \quad \text{where } k \text{ is a positive constant}$	5	3
Q 4	Attempt any 2 parts of the following		
4a	Define Wide Sense Stationary Process. Consider $X(t) = A \cos(\omega t + \theta)$. Is $X(t)$ a WSS process for the following cases? Justify your answer. (i) A and ω are constants and θ is a uniform random variable over $(-\pi, \pi)$. (ii) ω and θ are constants and A is a random variable.	1+2+2	4
4b	(i) Define Gaussian Random Process and mention at least 4 properties. (ii) Write down at least four properties of Auto correlation and cross-correlation function	5	4
4c	(i) Define Ergodic Random Process. Given $X(t) = A \cos(\omega t + \theta)$. A and ω are constants and θ is a uniform random variable over $(-\pi, \pi)$. Is $X(t)$ ergodic in both mean and autocorrelation? Justify your answer. (ii) Let $X(t) = A \cos(\omega t) + B \sin(\omega t)$ and $Y(t) = B \cos(\omega t) - A \sin(\omega t)$. A and B are independent random variables both with zero mean and variance σ^2 and ω is a constant. Find $R_{XY}(\tau)$.	3 2	4
Q 5	Attempt any 2 parts of the following		
5a	(i) Distinguish between White Noise and Band-Limited White Noise with respect to their Power Spectral Density and Autocorrelation Function. (ii) Consider an ideal bandpass filter with frequency characteristics as below:  The input to this filter is a white noise process. Determine the total noise power at the output.	3 2	5
5b	Proof that if a WSS random process passing through an LTI system the output random process is also a WSS random process using suitable mathematical expressions.	5	5
5c	(i) A WSS random process $X(t)$ with autocorrelation function $R_X(\tau) = e^{-a \tau }$, where a is a real positive constant, is applied to the input of an LTI system with impulse response $h(t) = e^{-bt}u(t)$, where b is a real positive constant. Find $R_Y(\tau)$ for output $Y(t)$. (ii) The input $X(t)$ to the RC filter shown below is a white noise process.  Determine the power spectrum and auto-correlation of $Y(t)$	2 3	5